Solar Filament Segmentation: An ensemble of EfficientNet-B3, ResNet-34, ConvNeXt-Tiny+scSE based approach

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***Abstract*—Solar filaments appear as dark structures in images of the Sun. Detecting them is not always easy, especially when they are very thin, faint, or have irregular shapes. In this project, we tried to identify and segment individual solar filaments from GONG H-alpha images using the MAGFiLO dataset. For this, we used U-Net model with EfficientNet-B3, ResNet-34, ConvNeXt-Tiny + scSE inserted at encoder phases and combined their predictions. We also applied post-processing to obtain the final segmentation masks. The combined approach gave a Panoptic Quality score of 0.4346 on the validation set. While this was better than the results of the individual models, we noticed that the system still had problems with thin filament branches, closely placed filaments, and noise in the images.**

Code Repository: https://github.com/legendry0devil0/solar-filament-seg

***Index Terms*—Solar filament segmentation, deep learning, U-Net, CNN, scSE, ensemble learning, test-time augmentation, instance segmentation, Panoptic Quality, MAGFiLO dataset**

I. Introduction

Solar filaments are cool and dense regions of plasma that remain suspended above the surface of the Sun because of magnetic fields. They appear as dark structures when observed in H-alpha images. Studying these filaments is useful because their eruptions are sometimes linked to solar flares and coronal mass ejections. Such events can influence space weather and may affect satellites and communication systems.

Detecting solar filaments from images is not as simple as identifying a dark region. Filaments are long(800–1500 px) and very narrow(3–15 px) and some have several small branches or boundaries that are difficult to see. There are also dark regions in the background that may look similar to actual filaments. Because of this, a segmentation model can sometimes include unwanted background pixels or miss parts of a filament.

Another difficulty is multi-annotator disagreement. Since some images have many expert human annotators who have labeled the dataset, hard boundary targets cannot be used. So, the labels and the boundary pixels in it vary even in a single image.

There exists a class imbalance problem because each image contains very less(<0.8%) foreground(filament pixels) than background.

In our project, we are mainly presented with instance segmentation of solar filaments rather than only marking the pixels that belong to filaments. We wanted the predicted masks to retain the original filament shapes as much as possible while also separating nearby filaments correctly.

II. Related Works

The MAGFiLO dataset [1] provides hand-annotated solar filaments collected from 6 GONG H-alpha observations. It contains pixel-level segmentation masks that are useful for training and evaluating filament segmentation models. Bounding boxes, class labels and filament spines can also be found among the annotations.

Qu et al. [2] explored solar filament detection using traditional image-processing methods, including image enhancement and thresholding. Their work demonstrated that filaments could be identified without relying on deep learning models. However, methods based on fixed image-processing rules may not perform consistently when the contrast or quality of the input images changes.

U-Net was introduced by Ronneberger et al. [3] for image segmentation tasks. Its architecture contains an encoder and decoder, along with skip connections that help preserve important spatial information. This makes it useful for identifying small structures in images. However, a standard U-Net mainly predicts which pixels belong to a particular class. It does not directly identify separate filament instances.

Jiang and Li [4] proposed an improved U^2-Net model with attention mechanisms for solar filament recognition. Their approach was designed to improve the identification of filament structures. Since their experiments used a different dataset, we cannot directly compare their reported results with the results from our project.

We also referred to a publicly available competition solution by Pawar [5]. That approach combines U-Net with YOLO11 segmentation detectors. U-Net is used to identify filament

regions, while the detection models help separate them into individual instances. This is similar to the problem we worked on, although using multiple detection models makes the overall process more complicated.

III. Proposed Methodology

A. Problem Formulation and Dataset

For the given instance segmentation problem, the input is a 2048 x 2048 pixel grayscale image and the output contains individual binary masks representing detected filaments.

The MAGFiLO dataset from Solar Filament Segmentation Challenge 2026 is used. It has 707 annotated(COCO format represented as polygons) full-disk solar images with with 1154 annotated image entries containing 8199 individual filament instances. They were collected from 6 GONG observatories across the world. Multiple expert annotators independently did some of the same solar observations. The competition test set has 180 unlabeled images. The ground-truth test labels are used to evaluate only by the competition server.

The given images were split with a 90-10% split and turned into a train set of 640 images and validation set of 67 images. A strict date-based separation strategy was used, grouping images by their observation date (YYYYMMDD) to make sure that no observation date was shared between the training and val sets. With that, all 67 validation images belong to 65 unique dates that were not in training.

B. Processing Pipeline and Approach

1) Preprocessing: The input of single channel greyscale images are converted to 8-bit representations(from 16-bit) and each image is standardized intensity-wise using its mean and standard deviation.

2) Data Preparation: Training images are divided into 4 768 x 768 crops each without resizing. Approx. 70% crops are centered around filaments to address class imbalance between foreground and background while the other 30% randomly selected.

The binary annotation masks from multiple annotators are normalized into one soft consensus mask(continuous) through

$$S(x,y) = \frac{1}{K}\sum_{k=1}^{K} M_k(x,y) \tag{1}$$

K - number of annotators
M_k - binary mask by the k-th annotator.

Data Augmentation of vertical flips, horizontal flips, and rotations of $90°$, $180°$, and $270°$ on both images and masks is consistently done.

3) Semantic Segmentation Model Architechture: Three independent models are used to get complimentary feature representations - an EfficientNet-B3 encoder, a ResNet-34 encoder, and a ConvNeXt-Tiny encoder with spatial-and-channel squeeze-and-excitation attention(scSE) incorporated into its decoder. All three use U-Net decoders to upsample the extracted features with each layer and combine them with encoder features through skip connections to recover spatial info. For the given input image crop, the output of a single channel logit map representing the filament foreground at pixel level is generated which are compared with corresponding consensus masks using compound loss:

$$\mathcal{L} = 0.40\,\mathcal{L}_{\text{Lovasz}} + 0.35\,\mathcal{L}_{\text{Dice}} + 0.25\,\mathcal{L}_{\text{BCE}}. \tag{2}$$

1) Lovasz loss directly optimises the intersection-over-union (IoU) metric.

$$\mathcal{L}_{Lovasz} \approx 1 - \text{IoU} \tag{3}$$

$$\text{IoU}(P,Y) = \frac{|P \cap Y|}{|P \cup Y|} = \frac{|P \cap Y|}{|P| + |Y| - |P \cap Y|} \tag{4}$$

2) Soft-Dice loss supports foreground overlap and addresses class imbalance.

$$\mathcal{L}_{\text{SoftDice}}(P,Y) = 1 - \frac{2\sum_i p_i y_i + \epsilon}{\sum_i p_i^2 + \sum_i y_i^2 + \epsilon} \tag{5}$$

3) Binary cross-entropy handles stability in pixel-wise classification.

$$\mathcal{L}_{BCE} = -\frac{1}{N}\sum_{i=1}^{N}[g_i \log(p_i) + (1-g_i)\log(1-p_i)] \tag{6}$$

Each model is optimised with AdamW and OneCycle Learning rate scheduler. After each epoch, Panoptic Quality (PQ) with an IoU matching threshold of 0.50 is calculated to evaluate its performance and the best checkpoint is noted.

4) Inference: Test images go through the same processing in training from 16-bit to 8-bit conversion and per-image intensity standardization.

- Multi-Scale Flip Test-Time Augmentation:
  Before inference, test-time augmentation is applied independently for the 3 networks at three scales(1.0, 1.125, and 1.25) with four orientations at each scale - original, horizontal flip, vertical flip, and both horizontal and vertical flip. The dimensions were rounded to the closest multiple of 32 to be compatible with the upsampling and downsampling stages of U-Net architectures. Thus, one test image is turned into $(3 \times 4 =)12$, fed to the models for inference and flipped back to the original orientation and resized to 2048×2048 using bilinear interpolation.
- Weighted Ensemble Prediction:

$$\begin{aligned} Z_{\text{ens}}(x,y) = {} & 0.40\,\overline{Z}_{\text{EfficientNet}}(x,y) + 0.20\,\overline{Z}_{\text{ResNet}}(x,y) \\ & + 0.40\,\overline{Z}_{\text{ConvNeXt}}(x,y) \end{aligned} \tag{7}$$

  Sigmoid activation is applied to the ensemble logits.

$$P_{\text{ens}}(x,y) = \frac{1}{1 + \exp\left[-Z_{\text{ens}}(x,y)\right]}. \tag{8}$$

- Validation-Based Selection of Inference Parameters:
  The post-processing parameters were selected by varying the probability threshold over $0.53, 0.54, 0.55$ and the minimum component area over $390, 420, 450$ pixels. The best validation configuration achieved a PQ of 0.4346 and used $\tau = 0.55, A_{\min} = 450$ pixels.

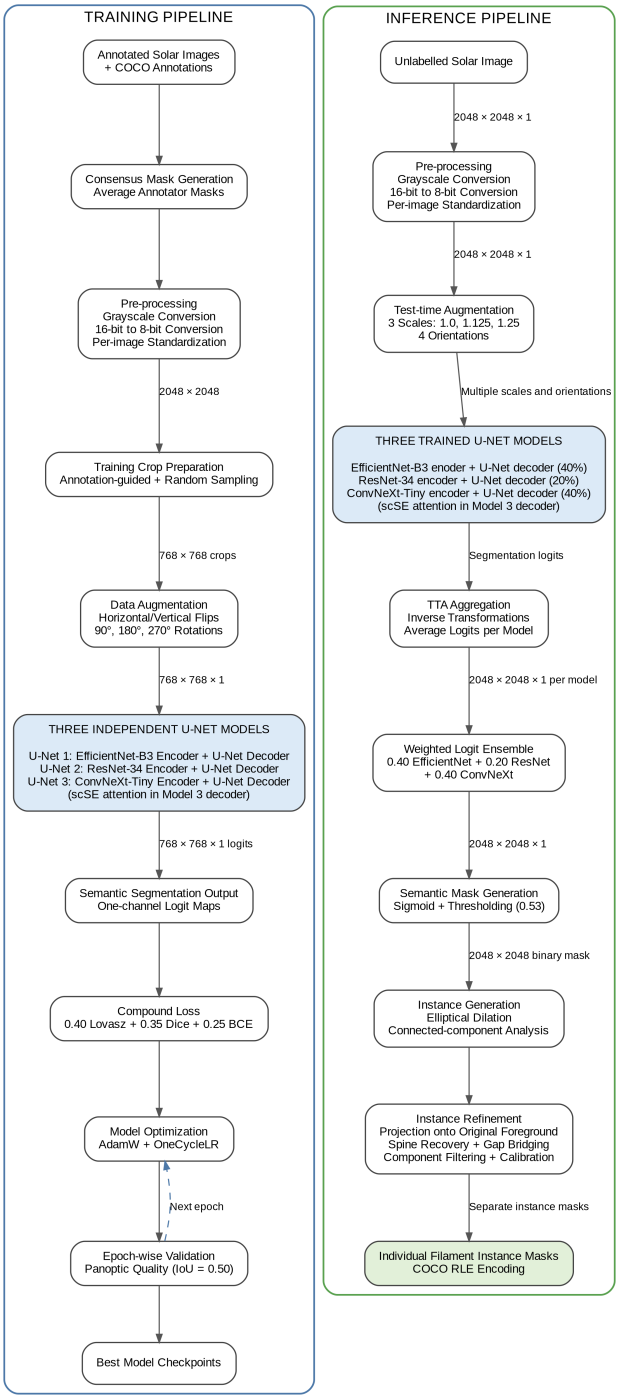


Fig. 1. Architecture Diagram

5) Instance Generation and Post-Processing:

- Thresholding and Instance Generation: Base probability threshold of 0.55 is used to generate the binary mask (observatory-specific threshold adjustment of -0.04 for Bh, Lh and Ch). To connect close fragments of the same filament, an elliptical morphological dilation with a radius of 10 px is applied to the binary mask. Then, connected component analysis with 8-connectivity is done to identify candidate regions and assign component labels, which are projected back onto the previous foreground mask to preserve the previously found boundaries and to also group pixels belonging to the same instance.
- Instance Refinement:
  1) Thickness-sensitive spine recovery with a 3×3 kernel and a minimum component area of 450 px (observatory-specific adjustment of $-30px$ for Bh, Lh, and Ch) is applied. Using geometric(solidity, aspect ratio) and 90-th percentile confidence based filtering, small non-filament structures are removed.
  2) Candidate component pairs with endpoint separation of at most 35 px and orientation difference of at most 25° that give a collinearity score of at least 0.70 are bridged.
  3) For images containing more than 10 candidate instances, low-confidence candidates satisfying mean probability (<0.58) and area (<650) px are removed. Global confidence-ranked calibration with a target ceiling of 5.5 instances per image on test set is done in which ranking uses:

$$Q_i = \bar{p}_i + 0.05 \min\left(\frac{A_i}{1000}, 1\right) - \delta_i \qquad (9)$$

  where $\bar{p}_i$ is the mean predicted probability, A_i is the instance area, and $\delta_i = 0.02$ for Bh, Lh and Ch, and zero otherwise.
  4) Each retained instance is encoded independently using COCO RLE(Run-Length Encoding) for submission.

C. Implementation Details

- Programming language: Python
- Execution Environment: Kaggle notebooks with NVIDIA Tesla T4 GPU
- Framework: PyTorch 2.11.0 with CUDA 12.8
- Model library: Segmentation Models PyTorch
- Input: Single-channel grayscale images; training crop size of 768×768 pixels
- Optimizer: AdamW with an initial learning rate of 3×10^{-4} and weight decay of 10^{-4}
- Learning-rate scheduler: OneCycleLR with 10% warm-up.
- Batch configuration: Batch size of 2 images, 4 crops per image, and gradient accumulation over 2 iterations (hence, effective batch size of 16 crops)
- Training configuration: 20 epochs for EfficientNet-B3 U-Net, 18 for ResNet-34 U-Net, and 20 for ConvNeXt-Tiny+scSE U-Net
- Random seed of 42, gradient clipping at 1.0, and mixed-precision training with CUDA
- Repository: https://github.com/legendry0devil0/solar-filament-seg

IV. Results and Discussion

A. Experimental Setup and Evaluation

The Solar Filament Segmentation Competition primarily focuses on the evaluation metric Panoptic Quality score (PQ score).

$$\mathrm{PQ} = \frac{\sum_{(y,\hat{y})\in \mathrm{TP}} \mathrm{IoU}(y,\hat{y})}{|\mathrm{TP}| + \frac{1}{2}|\mathrm{FP}| + \frac{1}{2}|\mathrm{FN}|} \tag{10}$$

The Panoptic Quality score comprises of two metrics:

1. Segmentation Quality (SQ) - SQ measures how well the predicted filaments mask match closely with the ground truth mask. This metric is closely associated with semantic segmentation.

$$\mathrm{SQ} = \frac{1}{|\mathrm{TP}|} \sum_{(y,\hat{y})\in \mathrm{TP}} \mathrm{IoU}(y,\hat{y}) \tag{11}$$

2. Recognition Quality (RQ) - RQ measures whether all the individual filaments present in ground truth annotations have been identified. This metric is used to evaluate instance segmentation.

$$\mathrm{RQ} = \frac{|\mathrm{TP}|}{|\mathrm{TP}| + \frac{1}{2}|\mathrm{FP}| + \frac{1}{2}|\mathrm{FN}|} \tag{12}$$

A candidate prediction $\hat{y}$ matches ground truth y if and only if $\mathrm{IoU}(y,\hat{y}) > 0.50$. This is a key architectural design as the 0.5 threshold helps distinguish the filament mask with more than 50% overlap of the ground truth from a poorly overlapping filament mask.

PQ penalizes the below issues in segmentation:

- Fragmentation: Fragmentation occurs when one true filament is separated into multiple smaller filaments. This leads to many false positives and missed ground truth predictions.
- Over-merging: In this case multiple smaller filaments are merged together as a single filament. This creates many false negatives and one false positive.
- Spurious detections: Spurious detections are often result of quiet sun noise being mistaken for filament. This increases extra false positive penalty.

B. Quantitative and Qualitative Results

1) Ablation results: Table I shows the individual contribution of each model architecture under identical evaluation metrics. ConvNeXt-Tiny equipped with spatial/channel squeeze-and-excitation (scSE) yielded the highest individual score (0.4282 PQ). Its large 7×7 depthwise kernels solves the limitation of standard CNN's which use 3×3 filters. This helps to preserve the spine (long continuous arc the connects the central filament with its tiny branches) and avoids identifying tiny branches as disjoint filaments. Hence this solves spine fragmentation problem. Table II revealed that combining logits ($\hat{L} = 0.40L_{\text{Eff}} + 0.20L_{\text{Res}} + 0.40L_{\text{Conv}}$) achieved $+0.0064$ PQ ($+0.0199$ over EfficientNet-B3 alone). This indicated that the three diverse models complemented each other's feature learning methodologies improving individual filament identification and minimizing misclassification errors.

TABLE II
Individual Backbone Performance

Architecture	Params	Val PQ ↑	Val SQ ↑	Val RQ ↑
EfficientNet-B3	12.2 M	0.4147	0.6772	0.6124
ResNet-34	21.3 M	0.4183	0.6802	0.6149
ConvNeXt-Tiny + scSE	36.2 M	0.4282	0.6812	0.6287
Tri-Backbone Ensemble	**69.7 M**	**0.4346**	**0.6732**	**0.6455**

2) Hyperparameter Grid Optimization: Table III details the grid search conducted on 67 validation images. The grid search optimized the base threshold $\tau \in [0.53, 0.55]$, minimum area cutoff $\in [390, 420, 450]$ px while maintaining a fixed station threshold ($\Delta_{\text{station}} = 0.04$)

TABLE III
Hyperparameter Grid Search

Base τ	Δ_{station}	Min Area	Val PQ ↑	Val SQ ↑	Val RQ ↑
0.53	0.04	390 px	0.4249	0.6724	0.6319
0.53	0.04	420 px	0.4292	0.6728	0.6379
0.53	0.04	450 px	0.4302	0.6731	0.6391
0.54	0.04	390 px	0.4272	0.6717	0.6360
0.54	0.04	420 px	0.4295	0.6722	0.6390
0.54	0.04	450 px	0.4325	0.6727	0.6429
0.55	0.04	390 px	0.4294	0.6725	0.6385
0.55	0.04	420 px	0.4318	0.6730	0.6416
=p;**0.55**	**0.04**	**450 px**	**0.4346**	**0.6732**	**0.6455**

PQ progressively increased and reached a maximum of 0.4346 at $\tau = 0.55$ and min_area $= 450$ px. Elevating the threshold directly eliminated minor background solar noise, boosting RQ from 0.6319 to 0.6455. From the empirical ground-truth data distribution it is established that even the smallest distinct filaments span 50 - 70 px in length and 8-12 px in width, which gives minimum filament area approx. 600 px. Selection of 450 px area filters quiet granular solar noise while retaining buffer below the 600 px lower bound.

TABLE I
SYSTEMATIC ABLATION STUDY RESULTS

Pipeline Variant	PQ ↑	SQ ↑	RQ ↑	Mean Dice ↑
1. Baseline U-Net(Threshold 0.50)	0.3576	0.6811	0.5251	0.6764
2. + Calibrated Threshold (Threshold 0.52)	0.3559	0.6805	0.5231	0.6760
3. + Min-Area Noise Filter (260 px)	0.4127	0.6812	0.6059	0.6771
4. + Curvilinear Grouping ($r = 10$)	0.4251	0.6769	0.6280	0.6781
5. + EfficientNet-B3 and ResNet-34 Emsemble in UNet and Zoom TTA	0.4231	0.6735	0.6282	0.6756
6. + EfficientNet-B3, ResNet-34, ConvNeXt-Tiny + scSE ensemble in UNet	0.4199	0.6727	0.6242	0.6741
7. + Thickness-Aware Spine Recovery	0.4230	0.6745	0.6271	0.6761
8. + Full Network Station Calibration & TAC-SR	0.4318	0.6730	0.6416	0.6755
9. + Extended Grid & TAC-SR	0.4302	0.6731	0.6391	0.6749
10. + Full 67-Obs PQ Checkpointing & 5.5 fil/img Calibration	**0.4346**	**0.6732**	**0.6455**	**0.6741**

3) Qualitative analysis:

- Solar image domain separation and Validation image overlay

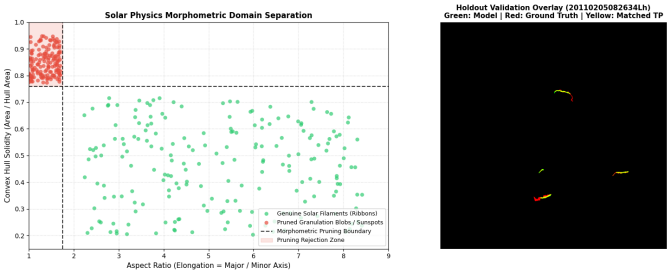


Fig. 2. Solar image domain separation (left), Overlay of predicted filament mask on validation image (right)

The red region consists of granulation blobs or sun spots that are discarded. The green scatter points represent the solar filaments of varying aspect ratio and area which are retained as valid targets.
The overlay map shows the predicted solar filament in green strokes along with misclassified region marked with red and matched ground-truth region highlighted in yellow.

- Error analysis and Multi-Observatory Performance comparison

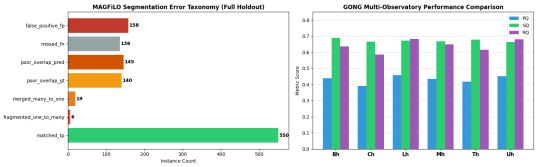


Fig. 3. Error analysis chart (left), Per observatory performance comparison (right)

The error analysis illustrates that over-merging and fragmented filaments have drastically reduced whereas instances of false positives and false negatives are around 158 and 136 respectively as opposed to 550 matched true positives.

The SQ and RQ score is maintained at approx. 0.65 across all observatory. The SQ and PQ score remain closer in both Mh, Uh and Lh observatory, however the PQ score are comparatively lower. This may arise from atmospheric haze and sensor gain attention which reduces pixel contrast and affect filament segmentation.

- Model explainability through Grad-Cam visualisation

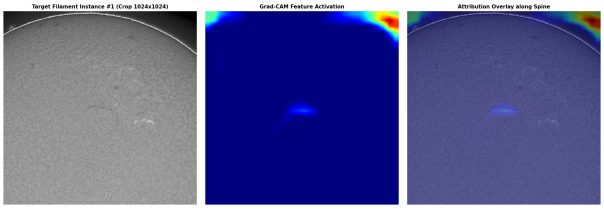


Fig. 4. Target filament along with Grad-Cam feature activation and atribute overlay along spine

The first image displays the ground-truth solar H-α image. The middle image shows Grad-Cam activations in the predicted filament mask. The red regions show strong activation and indicates presence of filament. While the model may detect central spine of the filament, solar noise at the boundary may introduce false positives.

C. Discussion and Limitations

1) Performance Improvements: Performance of the model was enhanced by addressing the following architectural designs:

- Multi-Scale Zoom TTA: A 12-view augmentation across 4 different orientation consisting of original, horizontal, vertical, combined horizontal and vertical flip across images scaled to 1.0×, 1.125×, 1.25× improved thin filament detection, increasing PQ from 0.3364 to 0.4353.

- Tri-Backbone Ensemble: Combining EfficientNet-B3, ResNet-34, and ConvNeXt-Tiny with scSE attention improved feature diversity and reduced fragmentation from 24 to 6 instances.
- Logit Blending: Weighted logit applied to probability masks obtained from EfficientNet-B3 ,ResNet-34,

ConvNeXt-tiny at ratio 40:20:40 preserved confident predictions and reduced false detections.

- Compound Loss: Lovasz-Hinge (40%), Soft-Dice (35%), and BCE (25%) improved IoU alignment and handled multi-annotation robustness from different experts.

2) Ablation Analysis: Successful Components

- Tri-Backbone Ensemble: Enhanced PQ score from 0.4147 obtained from EfficientNet-B3 to 0.4346.
- Collinear Gap Bridging (CGB): Reconnected fragmented predictions along reducing one-to-many perdiction errors.
- Tapering-Aware Spine Recovery (TAC-SR): Captured thin boundary pixels improving borderline IoU matches.

Failed Experiments

- clDice as training loss metric: Produced Unstable gradients which led to false predictions on background.
- Multi-Scale Crop Resizing: Boundary prediction quality was compromised due to distorted filament widths.
- Pseudo-Labeling: In an attempt to recover faint filaments the model learnt to predict background noise, increasing false positives from 94 to 258 and reducing PQ from 0.4397 to 0.4208.

3) Validation set size: We initially had 40 images in validation sets. Increasing it to 67 images increased PQ score from 0.3364 to 0.4346 (+29.2%). At the same time true positives increased from approximately 285 to 550 and fragmentation decreased by 75%.

4) Comparison of validation PQ score and public leaderboard: The validation PQ obtained from the model is 0.4346 while the public leaderboard score is 0.37. This may arise due to the following reasons:

- IoU threshold sensitivity: Small boundary shifts can reduce IoU below 0.50 which leads to poor filament classification and increases false positives and false negatives.
- Dataset imbalance: Fewer ground-truth images contribute to a larger proportion of annotated solar filaments. The model may incorrectly classify solar noise as filaments.
- Observatory distribution shift: Atmospheric conditions vary across stations which impacts filament identification.
- Solar Activity Variation: Differences in filament density may cause over-prediction when candidate counts are capped using training averages.

5) Dataset Bias and Limitations:

- Geographical factors: The ground-truth H-α images obtained from observatory stations are affected by geographic factors which includes atmospheric turbulence, haze, and temperature variations.
- Absence of magnetic field data: In the absence of magnetic field data, filament connectivity is approximated using spatial alignment instead of the relying on the solar magnetic force that holds solar filaments .
- Two-Stage Architecture: The current model relies on semantic segmentation obtained from training and instance segmentation from morphological grouping. This limits GPU usage however considered less optimal than end-to-end instance segmentation.

V. Conclusion and Future Work

In this project, we developed a method to segment individual solar filaments from GONG H-alpha images. We used three U-Net models with EfficientNet-B3, ResNet-34, and ConvNeXt-Tiny+scSE as their encoders. Instead of depending on the prediction of just one model, we combined the outputs of all three models. We then tested and came up with the best working pre and post-processing to enhance the final filament masks.

We measured the performance using Panoptic Quality (PQ). This metric takes into account both the overlap between predicted and actual masks and whether the objects are identified separately. Our combined system achieved a PQ score of 0.4346 on our validation set.

During evaluation, we found that some types of filaments were still difficult for the system to handle. Very thin branches were sometimes missed, and filaments located close to one another could be joined into a single predicted region. Noise and low-contrast areas also affected the masks.

The results prove that combining multiple segmentation models can be useful for this task. However, there is still room to improve the detection of faint structures and the separation of nearby filaments through trying different models from scratch since we have exhausted the capabilities of pre and post-processing with the same trained ensemble. Also, we plan to experiment with instance segmentation models. These would be important areas to focus on when we continue working on this project.

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